Tingyu Zhang

tingyu.z@northeastern.edu. 206-741-8937. LinkedIn. GitHub. Seattle. Open to Relocation

EDUCATION

Northeastern University Seattle, Washington
Master of Computer Science, (GPA: 4/4) Core Courses: SWE, Distributed Sys. *September 2025 – May 2027*

Southern University of Science and Technology Shenzhen, Guangdong
Bachelor of Applied Mathematics(GPA:3.53/4) Core Courses: Linear Algebra, Statistics. *August 2021 – July 2025*

SKILLS

Programming Languages: Python (proficient in data analysis and GenAI training), Java (proficient in system design), MATLAB, C/C++, JavaScript/TypeScript, PostgreSQL, MongoDB, R, LaTeX.

Developer Tools: VS Code, PyCharm, IntelliJ, DataGrip, Docker, Kubernetes, Maven.

Libraries&Framework: Pandas, NumPy, Matplotlib, Pytorch, Tensorflow, Cupy, scikit-learn, seaborn, LangChain, React, Node.js, Spring Boot, Swift.

RESEARCH EXPERIENCE

LLM Agent Implementation and Fine-Tuning in Data Science February 2025 - May 2025

- Developed **prompt engineering pipeline** implementing data cleaning, normalization, and serialization to **JSON** format with accompanying visualization generation, building a 500+ example prompt corpus with interpolation curve ground truth labels for **supervised few-shot learning** and model benchmarking.
- Implemented LLM agent onto **SSH** to train the classification **vision-Transformer** and fine-tuning by **LoRA**, achieved 90.4% accuracy for astronomical data.
- Generated and trained **Deep Learning LeNet-5 neural network** on MNIST dataset (60K images) achieving 80.3% test accuracy to establish a benchmark classifier.

Remote GPU Server Implementation and Numerical Computation July 2023 - September 2023

- Develop Python code to realize the FFT matrix acceleration algorithm on a remote server with **GPU** using **pytorch**, **tensorflow** and **cupy**, achieved approximately 20% performance improvement.

PROJECT EXPERIENCE

AI-Powered Job Search Agent with Ollama *January 2025 – Present*

- Developed **autonomous job search agent** using **Ollama** with **Llama 3.1** model running locally, implementing **RAG** architecture with **ChromaDB** vector store for personalized job matching.
- Engineered **multi-agent system** for resume tailoring, cover letter generation, and application tracking using **LangChain** and **Python**, implementing **prompt chaining** and **few-shot learning** patterns that reduced application preparation time from 2 hours to 15 minutes per job.
- Built **web scraping pipeline** using **Selenium** and **BeautifulSoup** to aggregate postings from job-searching webs, integrated with **SQLite** database and **Celery** task queue for asynchronous processing, collecting 2K+ daily job listings with 88.3% data accuracy. May use multi-agents sychronizely to optimize.

AWS Microservices Platform Engineer *September 2025 – December 2025*

- Architected **cloud-native microservices** on **AWS ECS** with **Docker containerization**, **Application Load Balancer (ALB)** for traffic routing, and **auto-scaling policies**, reduced the develop time to 5-10 minutes
- Integrated **RabbitMQ** message queues with **Apache Commons Pool** for connection pooling, implementing manual acknowledgements and retry logic to ensure reliable **asynchronous processing**.

SpringBoot FullStack Book Management Web Application *November 2025 – December 2025*

- **Frontend**: Architected full-stack application using **React 19** with **TypeScript** and Tailwind CSS for the frontend, **Spring Boot** with **PostgreSQL** for the backend.
- **Backend**: Developed **RESTful APIs** with **Spring MVC** supporting full CRUD operations, including multipart file uploads for book covers with client-side image preview using the FileReader API.

PROFESSIONAL EXPERIENCE

Data Engineering Intern *July 2022 - August 2022*
HAOCHENG TECH HOLDING Shenzhen, Guangdong

- Architected and deployed **PostgreSQL** database schema using **DataGrip**, implementing B-tree and composite indexes on video metadata fields. Achieved a **15x performance improvement** by reducing search operation latency from 30 minutes to 2 minutes through index placement.
- Engineered end-to-end data pipeline and developed classification models integrating classical **machine learning** algorithms (**Random Forest and Logistic Regression**) to predict material classification on a large csv dataset, achieving almost 87.3% accuracy.

Publications & Preprints

[1] T. Zhang, Z. Hu, and F. Kong, "Best-of-Three-Worlds Analysis for Dueling Bandits with Borda Winner," in *the International Conference on Learning Representations (ICLR)*, 2026.

[2] K. Zhou, T. Zhang, W. Chen, and F. Kong, "Hybrid Combinatorial Multi-armed Bandits with Probabilistically Triggered Arms," arXiv preprint arXiv:2512.21925, 2025.